Code of Professional Ethics

for AI Developers in Reproductive Embryo Selection Systems

Including IVF Embryo Ranking and Preimplantation Genetic Testing (PGT)

VERSION	DATE	FRAMEWORK
Draft 8	April 2026	Based on ABIM Physician Charter

ABSTRACT

This Code of Professional Ethics defines enforceable duties for artificial intelligence developers who design, validate, and deploy embryo selection systems, polygenic risk-scoring algorithms, and related machine learning tools for in vitro fertilization (IVF) and preimplantation genetic testing (PGT). Drawing on the ABIM Physician Charter and reproductive ethics literature, this Code establishes foundational professional commitments, lifecycle obligations, prohibited practices, and accountability provisions to ensure that AI-enabled innovation in reproductive medicine advances patient welfare, scientific integrity, and equitable access.

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I. Preamble

This Code of Ethics is for artificial intelligence (AI) developers who design and deploy embryo selection/ranking and polygenic risk-scoring machine learning systems. Similar to the professional oaths of service and bioethical principles that bind physicians, this Code outlines the professional duties of AI developers involved with in vitro fertilization (IVF) and preimplantation genetic testing (PGT).

While these AI developers do not practice medicine, their machine learning algorithms can directly influence clinical decisions about which embryos are transferred, and which are discarded. Bridging the gap between clinician-specific ethical frameworks and the unregulated influence of AI developers, this Code defines commitments, obligations, and prohibited practices for AI developers operating in this domain.

Scope of Application: Medical vs. Non-Medical Trait Selection

AI-assisted embryo selection systems enable prospective parents to compare embryos across two fundamentally different categories of traits, each carrying distinct ethical implications:

Medical Trait Selection	Embryos are screened and ranked based on predicted disease risk for clinically recognized conditions, including monogenic conditions (e.g., cystic fibrosis, Huntington's disease) and polygenic conditions (e.g., type 2 diabetes, hypertension, Alzheimer's disease).
Non-Medical Trait Selection	Embryos are ranked based on predicted non-disease traits associated with social or aesthetic preference such as eye color, hair color, height, and intelligence quotient (IQ).

This Code addresses these two categories, and it is pivotal that systems prioritize the risk of serious genetic disease or medical trait selection over those developed for non-medical, social, or aesthetic trait optimization. The inclusion of social traits is not categorically prohibited, but any such traits shall be explicitly identified, justified, and constrained, and shall not supersede or obscure medically relevant

considerations. Where national or regional laws prohibit screening for non-medical or social traits, developers shall comply with applicable regulations within those jurisdictions. In jurisdictions where such screening is legally permitted, the full obligations of this Code apply.

This framework was derived from the American Board of Internal Medicine (ABIM) Physician Charter and existing reproductive ethics literature. To ensure that AI-enabled innovations in IVF and PGT are deployed ethically and safely, AI developers must subscribe to fiduciary standards of patient welfare, scientific integrity, and social responsibility.

II. Foundational Professional Commitments

These six commitments define the core professional duties of AI developers involved with IVF and PGT, setting the ethical foundation from which all lifecycle obligations and prohibitions derive. They are modeled on the core principles of medical professionalism.

1. Responsible Embryo-Selection System Design

Developers shall place the welfare of patients and future children above all competing interests. No commercial or organizational pressure may override the developer's responsibility to avoid foreseeable harm for prospective parents and future children. While designing embryo-selection systems, developers shall prioritize patient safety and clinical utility over predictive scope and model complexity.

2. Transparent Presentation of Embryo Risk Scores

Developers shall present all polygenic risk scores and embryo rankings as probabilistic estimates, not deterministic predictions. No embryo risk score shall be provided to clinical partners without accompanying documentation of its predictive limitations. All system outputs provided to clinical partners shall be accompanied by confidence intervals, predictive validity across populations, and relevant reproductive outcome benchmarks, where available. In instances where proprietary data constraints limit disclosure, developers shall clearly identify what information has been withheld and why.

3. Support for Informed Reproductive Counseling

Although obtaining informed consent is the duty of the clinician, developers shall be responsible for ensuring that the materials and systems they provide make accurate counseling possible. Developers shall design systems and produce materials that enable genetic counselors and other designated professionals to accurately interpret and communicate AI-generated outputs.

At minimum, developers shall provide:

a) Counselor-facing training materials explaining model logic, confidence levels, and known limitations

b) Patient-facing documentation clarifying when AI systems are applied and what each output represents.

4. Equity in Polygenic Prediction

Developers shall assess and document how their systems perform across patient populations, with particular attention to ancestry-related disparities in predictive validity. Any system that demonstrates materially unequal predictive performance across populations shall not be deployed unless such disparities are disclosed and deemed acceptable within the clinical context. To mitigate health inequities, developers shall evaluate pricing structures, clinic partnerships, and eligibility criteria.

5. Scientific and Technical Integrity

Developers shall ground their work in peer-reviewed methodologies and established engineering practices. All systems shall undergo pre-deployment technical validation, including benchmarking against clinical comparators, evaluating performance across populations, and assessing model stability. Developers shall clearly communicate the probabilistic nature of all outputs and the limits of available evidence, including where clinical utility has not been established.

Although the collection and interpretation of long-term clinical outcomes remain within the clinical domain, developers shall design systems that support integration with such data for ongoing evaluation. Where new evidence materially affects the model's performance or interpretation, developers shall update, recalibrate, or suspend future deployment of the system, as appropriate.

6. Independence from Fertility Market Pressures

Developers shall proactively identify potential conflicts of interest between themselves and employers, clinic partners, or other parties that could compromise embryo-ranking criteria or system design. Developers shall ensure that identified conflicts do not bias the design, validation, or deployment of AI systems. Commercial relationships shall be documented and disclosed to clinical partners prior to deployment.

III. Obligations Across the AI Development Lifecycle

The following obligations are organized by phase of the AI development pipeline. Each phase defines what developers are directly responsible for and distinguishes those responsibilities from obligations that belong to clinical partners.

A Data and Model Design

Developers shall build training datasets that are representative across patient populations and evaluated for ethical consequences (e.g., biases or inequities). Dataset composition, including ancestry distribution and clinic source, must be internally documented prior to model training.

Trait inclusion shall be explicitly justified and documented. Traits related to the prevention of serious genetic disease or clinically meaningful reproductive outcomes shall be prioritized. Where non-medical traits are included, developers shall:

- **a)** Clearly distinguish such traits from clinical indicators
- **b)** Provide written justification for their inclusion grounded in transparency and clinical context
- **c)** Ensure that non-medical traits do not supersede or obscure medically relevant information.

AI models shall be designed with interpretability as a core requirement, including confidence intervals and output explanations for clinical use.

B Validation and Testing

Validation of the developer's own system shall follow a phased evaluation process consistent with the general principles underlying the FDA's staged pre-market approval framework for medical devices. Validation must start with early feasibility studies through multi-site clinical testing and progress to real-world effectiveness assessment. Developers are not responsible for validating systems developed by other parties.

Developers shall be responsible for pre-deployment technical validation, including:

- **a)** Benchmarking against prior embryologist decisions or established clinical comparators
- **b)** Bias audits across relevant population subgroups
- **c)** Stability testing to assess output consistency under dataset shifts and retraining

If model outputs vary significantly across dataset shifts or retraining, deployment shall be deferred until stability is established. Iterative deployment does not exempt developers from meeting minimum performance and safety thresholds before clinical integration.

As developers lack the patient access and longitudinal follow-up capacity required for long-term clinical outcome validation, clinical partners shall oversee the collection of specific performance metrics (e.g., live birth rate, implantation rate, miscarriage rate, trait-related outcomes). Where authorized data-sharing agreements exist, developers shall support clinical partners in conducting this evaluation.

C Deployment and Use

Developers shall require and document role-specific training for all end users—including clinicians, embryologists, genetic counselors—prior to clinical deployment. Training shall address interpretation of uncertainty, contraindications, and common patterns of misuse.

Deployment documentation shall clearly define:

- **a)** The system’s intended use cases
- **b)** Known limitations and contraindications
- **c)** Conditions under which the system should not be used

Systems shall be deployed with an appropriate human oversight model. Because human review can either reduce or increase error depending on task complexity, developers shall evaluate empirically whether human-in-the-loop review improves outcomes for the intended use-case and shall implement oversight accordingly.

No AI model shall be deployed for clinical use without meeting applicable regulatory requirements and internal standards for safety, performance, and transparency. Marketing and presentation must not overstate the model's predictive capabilities.

D Monitoring and Maintenance

Developers shall implement tools and processes to monitor performance drift post-deployment. Usage log review protocols and monitoring responsibilities shall be established and documented before deploying these models. Where developers retain access to post-deployment data, usage logs shall be reviewed on a regular and pre-specified schedule for evidence of discriminatory effects or unintended clinical consequences.

When minimum performance or equity thresholds are no longer met, developers shall notify clinical partners promptly and either recalibrate or suspend the system until thresholds are restored. All corrective actions taken shall be communicated to clinical partners in a timely manner.

IV. Prohibited Practices

The following practices are strictly prohibited under this Code.

1. *Developers shall not design embryo-ranking systems in which social or non-medical traits are prioritized over the prevention of serious genetic disease or clinically meaningful reproductive outcomes.*

2. *Developers shall not represent probabilistic polygenic risk scores or embryo rankings as deterministic predictions or clinical guarantees.*

3. *Developers shall not market or clinically integrate embryo-selection systems without documented validation criteria and pre-specified performance benchmarks.*

4. *Developers shall not conceal known limitations in training data composition or population-specific predictive gaps from clinical partners.*

5. *Developers shall not conceal evidence of model bias, significant performance degradation, or safety-relevant system failures from clinical collaborators.*

6. *Developers shall not label systems as "research only" while knowingly facilitating their use in clinical decision-making.*

7. *Developers shall not independently market embryo-selection AI tools directly to patients without physician mediation and appropriate regulatory oversight.*

V. Accountability, Documentation, and Oversight

The following provisions establish the structural conditions under which developers remain accountable for the systems they build and deploy.

1. Developers shall maintain comprehensive documentation of model design decisions, training data composition, validation procedures, performance benchmarks, and update history. Documentation shall be sufficient to enable retrospective review of system behavior, including evaluation of model performance, limitations, and decision pathways.

2. Developers shall communicate material system limitations and performance concerns to clinical collaborators in a timely manner. When a system's performance degrades below established thresholds, developers shall notify clinical partners before continued clinical use.

3. Developers shall work with institutional and clinical partners to clearly define contractual boundaries of responsibility regarding system interpretation and clinical use. These boundaries shall be established prior to deployment and documented in a form accessible to all parties.

4. Developers shall seek interdisciplinary input—including from reproductive specialists, genetic counselors, and bioethicists—when making decisions that expand trait scope, modify embryo-ranking criteria, or alter the clinical use conditions of deployed systems.

VI. Conclusion: Ethical Stewardship in Reproductive AI

AI developers who design embryo-selection and polygenic risk-scoring systems bear professional responsibility for the ethical consequences of their work. Although they do not practice medicine, their systems shape which embryos are prioritized, transferred, or foregone: decisions with lasting consequences for patients, families, and resulting offspring. The principles, obligations, and prohibitions set forth in this Code define the ethical boundaries of that role.

Adherence to this Code requires sustained commitment to scientific rigor, transparent communication of uncertainty, equitable performance across populations, and independence from commercial pressures. It also requires restraint: systems shall not be deployed until adequate validation is complete, and predictive claims shall not exceed the strength of available evidence.

This Code does not assign clinical obligations to developers, nor does it place them in the role of clinicians. It defines the professional responsibilities that arise from the developer's technical authority over system design, validation, deployment, and post-deployment conduct.

As the field evolves, this Code shall be subject to periodic review and revision through structured stakeholder engagement and professional society oversight. Professional responsibility in reproductive AI is not optional. It is a condition of legitimate innovation in reproductive medicine.